

Wallet System: ER Diagram & User/Admin/Settlement Flow

Laravel project **/fundraise** — generated from source models and services.

Entity Relationships

From	Relation	To	Notes
User	1 : 0..1	Wallet	owns via owner_type=User / owner_id
Organization	1 : 0..1	Wallet	owns via owner_type=Organization / owner_id
User	1 : 0..*	Organization	owner (user_id)
User	1 : 0..*	Campaign	creator (user_id)
Organization	1 : 0..*	Campaign	runs
Campaign	1 : 0..*	Donation	receives
Donation	1 : 0..*	WalletTransaction	credit source=donation
Donation	1 : 0..1	SettlementItem	allocated to
Donation	1 : 0..1	Refund	may have
Wallet	1 : 0..*	WalletTransaction	ledger
Organization	1 : 0..*	CampaignSettlement	requests payout
Organization	1 : 0..*	PayoutAccount	has
CampaignSettlement	1 : 0..*	SettlementItem	contains
User	1 : 0..*	CampaignSettlement	approved_by / rejected_by (admin)

Entity Attributes

WALLET

Field	Type / Key	Description
id	PK bigint	primary key
user_id	FK bigint	legacy owner user
owner_type	string	User Organization (morph)
owner_id	bigint	morph id
balance	decimal(2)	available / withdrawable
reserved_balance	decimal(2)	hold window (7 days)
pending_settlement_balance	decimal(2)	locked in payout
currency	string	INR

WALLET_TRANSACTION

Field	Type / Key	Description
id	PK bigint	primary key
wallet_id	FK bigint	-> wallet
type	string	credit debit
amount	decimal(2)	signed magnitude
source	string	donation refund settlement adjustment gift_card coupon
reference_type	string	Donation Refund CampaignSettlement ...
reference_id	bigint	idempotency key
balance_after	decimal(2)	snapshot

status	string	completed
notes	text	reason / audit

CAMPAIGN_SETTLEMENT

Field	Type / Key	Description
id	PK bigint	primary key
campaign_id	FK bigint	-> campaign
organization_id	FK bigint	-> organization
gross_amount	decimal(2)	sum total_amount
platform_fee	decimal(2)	sum platform_fee
net_amount	decimal(2)	sum net_amount
status	string	pending_approval paid rejected failed
approved_by	FK bigint	admin user
rejected_by	FK bigint	admin user
rejection_reason	text	required on reject
paid_at	datetime	when payout sent
gateway_reference	string	payout reference

SETTLEMENT_ITEM

Field	Type / Key	Description
id	PK bigint	primary key
campaign_settlement_id	FK bigint	-> settlement
donation_id	FK bigint	-> donation
amount	decimal(2)	= donation.net_amount

DONATION

Field	Type / Key	Description
id	PK bigint	primary key
user_id	FK bigint	donor / creator
campaign_id	FK bigint	-> campaign
total_amount	decimal(2)	gross
platform_fee	decimal(2)	fee
net_amount	decimal(2)	net to wallet
payment_status	string	completed
is_refunded	bool	refund flag
paid_at	datetime	hold window start
released_at	datetime	reserve matured stamp

ORGANIZATION / USER / PAYOUT_ACCOUNT

Field	Type / Key	Description
organization.id	PK	user_id owner, type, verification_status, wallet_hold_days
user.id	PK	name, email (wallet owner / admin)
payout_account.id	PK	organization_id, is_verified, account_ref

Donation Money Lifecycle

reserved_balance → balance → pending_settlement_balance → paid

1. **Donation credited** → lands in **reserved_balance** (7-day fraud/refund hold).
2. **Daily cron** wallet:release-reserves moves matured funds reserved- → balance+.
3. **User requests payout** → balance- → pending_settlement_balance+ (locked, not debited).
4. **Admin approves** → pending_settlement-, debit SOURCE_SETTLEMENT, payout initiated.
5. **Admin rejects** → funds returned pending_settlement- → balance+.

User Side vs Admin Side

Stage	User / Org Side	Admin Side
Wallet	getOrCreateWallet on first use (INR, 0)	View all wallets, per-owner ledger
Receive	Donation webhook -> credit (reserved)	(automatic)
Release	Cron matures reserve -> balance	(automatic)
Payout	Dashboard selects eligible donations -> requestPayout	Settlement list + scrutiny flags
Decision	(waits)	approve -> payout / reject -> funds back
Adjust	n/a	Manual credit/debit with notes
Refund	n/a	PaymentController debits wallet (reserved first)

Design Notes & Gaps

- All mutations run in DB::transaction with Wallet::lockForUpdate() (pessimistic locking).
- Credit/debit are idempotent on (wallet_id, reference_id, reference_type, source) - safe for webhook retries.
- One wallet per fundraiser: both donation credit and settlement debit use the same wallet owner.
- initiatePayout() is a STUB - logs a PAYOUT_* reference; no real Razorpay/Paytm disbursement yet.
- wallet_hold_days on Organization is NOT honored by the cron - it uses the hardcoded DEFAULT_HOLD_DAYS=7.
- CampaignSettlement.transfer_reference is declared but never populated (gateway_reference is used instead).
- No payout webhook reconciliation to flip status -> failed.